

Implement a Smile detection algorithm using Open CV to detect and locate human smile in the images.

Code:

```
import cv2, urllib.request
from google.colab.patches import cv2_imshow

# Download sample image
urllib.request.urlretrieve(
    "https://raw.githubusercontent.com/opencv/opencv/master/samples/data/lena.jpg",
    "smile.jpg"
)

# Download Smile Cascade file
urllib.request.urlretrieve(
    "https://raw.githubusercontent.com/opencv/opencv/master/data/haarcascades/haarcascade_smile.xml",
    "smile.xml"
)

# Read image
img = cv2.imread("smile.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Load classifier
smile_cascade = cv2.CascadeClassifier("smile.xml")

# Detect smiles
smiles = smile_cascade.detectMultiScale(gray, 1.8, 20)

# Draw rectangles around smiles
for (x, y, w, h) in smiles:
    cv2.rectangle(img, (x, y), (x+w, y+h), (255,0,0), 2)

# Display result
print("Smile Detection")
cv2_imshow(img)
```

Output:

mile Detection

